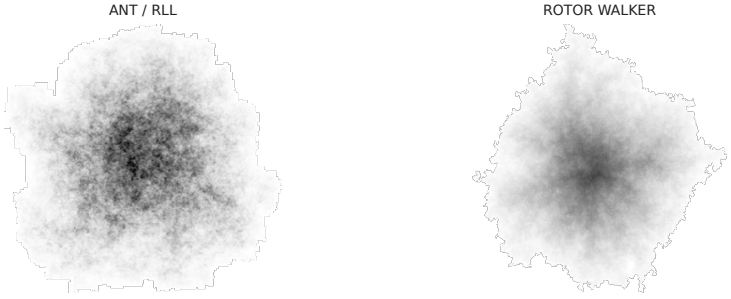
Cube-root growth of most non-highway generalised Langton's ants

Rules of length ≤ 12: a calibrated mean-field decomposition

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Abstract. A generalised Langton's ant follows a string over {L, R}: on a cell of colour c it turns by `rule[c]`, advances the cell to colour $(c+1) \bmod n$, and steps forward. Most rules never build a highway; they grow an irregular, roughly round territory. Does it grow like the rotor walker, whose radius grows as $t^{\frac{1}{3}}$ [1]? For most of the 3055 non-highway rules of length 3–12 that we simulated (to 2^{27} steps, six to 2^{32}), yes: the bounding-box width grows as t^{α} with median $\alpha = 0.333$, good to about ± 0.01 after calibrating the finite-window bias on the rotor, and the time-averaged visit density is the rotor's cone. Some rules do not follow it (425 grow logarithmically, 103 faster), and the edge does not match: the ants' surface width grows about twice as fast as the rotor's. Splitting the growth rate into an edge-occupation ratio κ and an edge-entry probability c (an identity, so not independent evidence for the cube root), we find that deviating rules deviate through κ, that c is predictable from the rule's letters (held-out $R^2 = 0.66$) and κ is not. What sets κ, and the ant's wider edge, is unexplained.



Same bulk growth. Different measured edge growth.
Occupation fields at 2^{22} steps; darkness shows visits per cell in units of $\frac{t}{A}$, on a shared scale from 0 to 4 (values above 4 clipped). Each territory is fitted to its panel.

Key results

- **Growth.** Over the 2622 blob rules the median width exponent is 0.333, with no dependence on rule length, and the area grows as $t^{\frac{2}{3}}$ to within the same bias (Section 4). The rule-bootstrap interval (0.332 to 0.334) resamples rules only and ignores a larger finite-window bias; individual rules differ by up to 0.02 at 2^{30} steps.
- **The bulk matches the rotor.** The time-averaged visit density is the cone $3(1 - r/R)$ to within 5% of the rotor's. The cone follows from uniform mixing plus $t^{\frac{2}{3}}$ growth, so this is a consistency check, not extra evidence (Section 7).
- **The edge does not.** The ants' surface width grows with an exponent of about 0.22–0.25 against the rotor's 0.10–0.13 (local exponents at matched sizes 2^{22}–2^{32}; a single fit from 2^{18} gives 0.26 against 0.16, and Kapri and Dhar report about 0.13). The gap does not close to 2^{34} steps (Section 7).
- **Not every rule obeys.** 425 rules grow logarithmically, 103 blob rules have $\alpha > 0.37$, and 304 more were excluded as builders or slow space-fillers. Highway classification is by finite simulation.
- **The decomposition.** Drift exponents below 0.05 hold for 99.7% of rules for c and 83% for κ, with an edge exponent of 0.52 in the census window. A drift of 0.05 shifts the area exponent by about 0.03, as much as the differences between rules (Section 5).
- **What is new.** Rules that violate the cube root do so through drift of κ, not of c. A chirality model explains 66% of the variance of c on held-out rules; nothing we tried predicts κ, which differs between rules moderately reproducibly (intraclass correlation 0.58) but is not carried by the letters, and a memoryless walker does not reproduce it, which motivates board-memory comparisons without proving memory sufficient (Section 5).
- **Unexplained.** The value of κ (which sets the prefactor), why it drifts in some rules, and the ant's wider edge. We say which parts are tautological, which are tested, and what remains open.

1 Introduction

Langton's ant [2] is a two-colour automaton whose behaviour is striking: after a chaotic transient of about 10 000 steps it builds a periodic "highway" and escapes. Generalising the turn rule to a string over {L, R} of length n [3], [4] gives a family whose long-time behaviour is, for most rules, unknown. For any rule containing both letters the track is known to be unbounded [3], [5], and the dynamics is computationally hard in general [6], yet we found no published measurement of how the territory of a rule that does *not* build a highway grows with time.

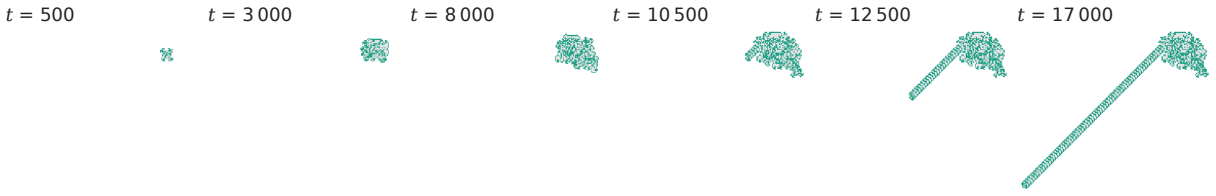


Figure 1: Langton's ant (RL) at six moments, all drawn at the same scale. For about 10 000 steps it wanders and builds an apparently random patch; then it falls into a repeating 104-step cycle that moves it two cells diagonally each time [2] (our sweep reproduces period 104 and onset 9976 for RL: `data/sweep_v2.csv`), laying a straight "highway" that continues indefinitely. The rules studied in this paper are those that never do this.

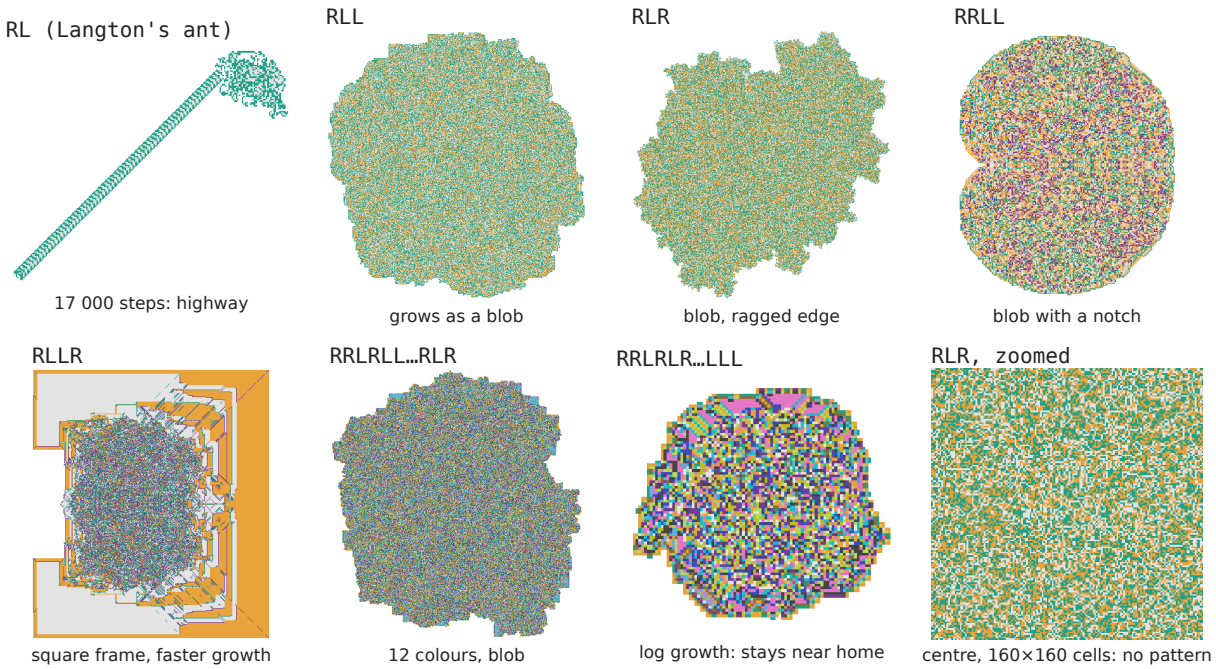


Figure 2: What the ants draw. Each colour is one of the n states a cell can be in; blank is never visited. Top left: classic Langton's ant (RL) after 17 000 steps, a tangle followed by a straight "highway" that runs off forever. The other panels are rules that never build a highway, after 2^{25} (33 million) steps: most grow into a roughly round blob whose texture looks random up close (bottom right), but the shapes differ (notched, ragged, or, for RLLR, a square frame) and one (RRLRLR…LLL) barely grows at all, staying near its starting point. Rule letters give the turn on each colour.

Here we make that measurement. The natural comparison is the rotor walker, in which every site holds an arrow that turns by 90° each visit and the walker follows it. For the rotor, radius $\sim t^{\frac{1}{3}}$ was conjectured by Priezzhev et al. [7], supported numerically by Kapri and Dhar [8], and proved by Bou-Rabee and Peres [1], in an arXiv preprint that we understand has not yet been refereed, for uniformly random initial arrows (range of order $t^{\frac{2}{3}}$, convex limit shape). The ant is not an Eulerian walker, so nothing implies it should match. We find that in the bulk it does: the exponents and the whole time-averaged occupation profile agree. At the edge it does not: the ants' surface width grows with roughly twice the temporal exponent, and we cannot tell whether that is a different universality class or a slow transient. We also find that the quantity that sets the growth prefactor (κ below) is a moderately reproducible property of each rule that no feature of the rule string predicts. These are the open problems; the calibrated census and the decomposition of the growth rate below are the tools that make them well posed.

Contributions.

1. A census of 3512 rules with growth exponents for 3055 of them, reporting the whole distribution, its dependence on rule length, and the rules that fall outside it.
2. A calibration: the same pipeline applied to the rotor walker (known exponent) and a random walk (negative control), giving the estimator's bias and between-seed noise.
3. A three-factor decomposition of the area growth rate, with the factors measured per rule, and a classification of deviating rules by which factor drifts.
4. A future-window forecast from early transport measurements, tested across replicated starts and against rotor and random-walk controls.
5. A held-out predictive model for the edge-entry probability from the rule's letters, and a test of the "well-mixed" assumption behind the second factor.

Section 5 and Section 8 separate what is an identity from what is a test, and list what we could not explain. A hypothesis about "homes" that tether some rules to their origin was investigated and is mentioned only in the limitations; we make no claim about it.

2 Model and methods

2.1 Rules, census and classes

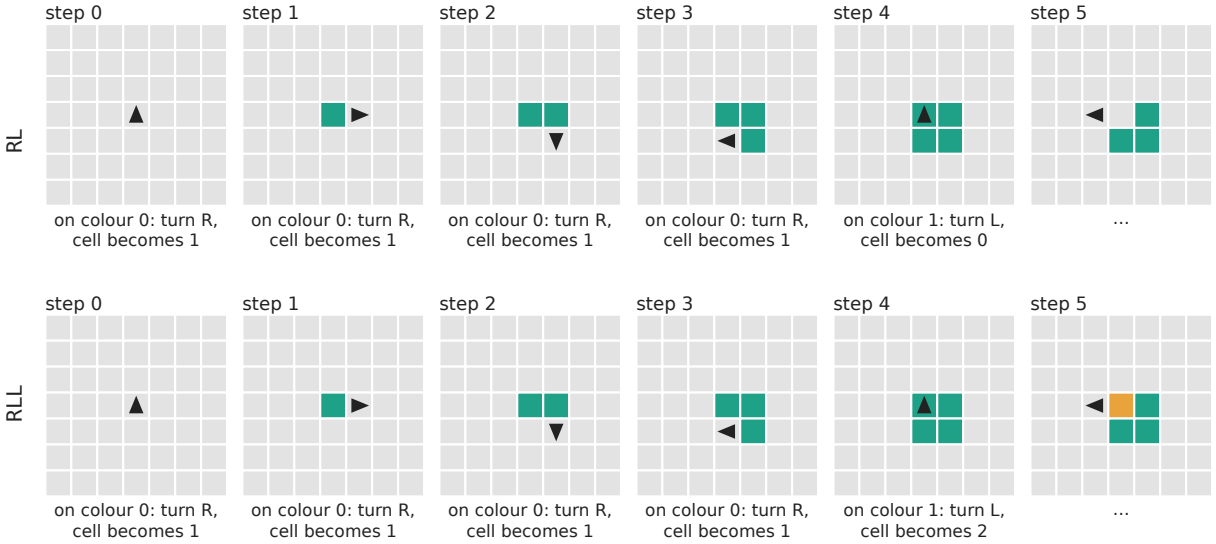


Figure 3: The first steps of two ants. The ant (black triangle) looks at the colour of the cell it stands on, turns left or right as the rule's letter for that colour says, changes the cell to the next colour (cyclically), and steps forward. RL has two colours, so a cell flips between blank and green; RLL has three (blank, green, orange).

A rule is a string $r \in \{L, R\}^n$. The ant has a position, a heading and a grid of colours in $\mathbb{Z}_n$, all initially 0. At each step it reads the colour c under it, turns left or right by `r[c]`, increments the colour to $(c + 1) \bmod n$, and moves one cell. Swapping every L and R gives the mirror-image ant, so we keep only rules starting with R, and drop the trivial rules in which every colour turns the same way.

A sweep of all such rules of length 2–12 (4083 rules) ran each on a blank 8192^2 grid for up to 10^8 steps with a highway detector (`data/sweep_v2.csv`); 578 escaped or were detected as highway builders (including classic RL). The census studied here is the remaining 3512 rules of length 3–12 (3505 that ran the full budget plus 7 that escaped late).

Each census rule was then run to 2^{27} steps (`data/growth.csv`, `data/census.py`). A rule is called a *builder* (304 rules) if it did not complete 2^{27} steps inside the grid or its steps per bounding-box cell at 10^8 steps were below 20; builders are excluded.

Most are not highway builders: the recognised highways were already removed above, only 5 of the 304 have a highway period in `data/sweep_v2.csv` (none confirmed), 289 stay in the grid and fill it slowly (about 2–6 steps per box cell, frame-like structures) and 10 left the grid (`data/b4_builders.txt`). We did not check that the 289 form one class.

Of the rest, a rule is *log-growing* (571 rules) if the fitted width exponent from $t = 2^{21}$ to 2^{27} is below 0.15, and otherwise *blob* (2637 rules). The cut at 0.15 is chosen by eye from the gap in the exponent histogram and has not been tuned against any outcome.

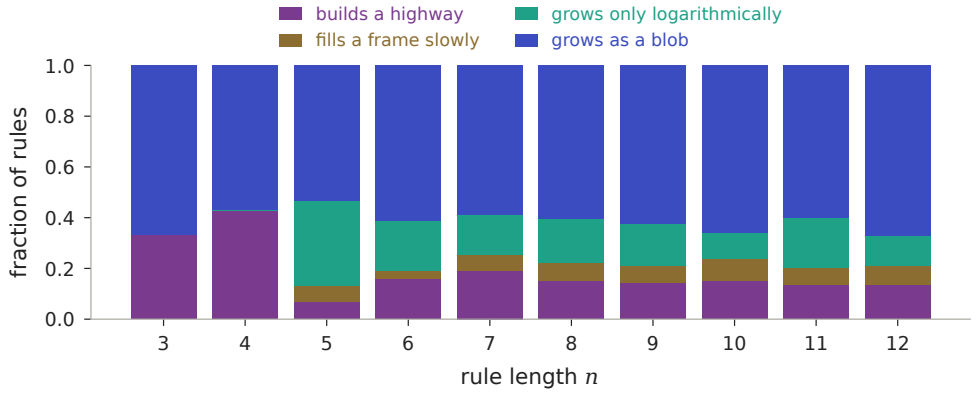


Figure 4: Outcome of every rule in the sweep (4083 rules of length 2–12; the single length-2 rule is RL and is not shown), by rule length. Highway builders are a third to two fifths of the length-3 and length-4 rules and about one in seven from length 6 on. "Frame" builders (mostly rules that fill a frame slowly, a few that leave the grid) are a small slice. The two top classes, blob and log-growing, are the subject of this paper.

2.2 Instrumented runs

The probe (`src/probe.rs`) tracks the *visited* set (every cell the ant has ever stood on), its area A, its *boundary* P (visited cells with at least one unvisited 4-neighbour), and, in octave windows $[2^k, 2^{k+1})$, the number of steps that start on a boundary cell and the number of steps that enter a never-visited cell. It also gives the bounding box, the radius of gyration R_g of the visited set, and the ant's own spread. For each rule we fit log–log slopes over windows $k = 20…26$:

- width exponent α (bounding-box side against t) and R_g exponent α_g;
- area exponent a ($A \sim t^a$);
- boundary exponent β ($P \sim A^\beta$).

3055 rules were analysed; 153 of the 3208 growing rules had no usable window set (a window contained no new cells) and were not fitted, and 8 were flagged as bursty and removed from the statistics below (the flag is a column of `data/probe_summary.csv` computed from the largest single-window jump, `max_jump`, and its share of growth, `max_share`; the script that set it was not preserved in the repository, but the flag is reproduced exactly by a threshold on `max_jump`: every flagged rule has `max_jump` ≥ 4.18 and every unflagged rule ≤ 3.59, and including the flagged rules moves the blob median width exponent from 0.333 to 0.333).

The unfitted rules are 5% of growing rules, and they are not fast growers: rerun on a 32768^2 grid, none left the grid, all reached 2^{26} steps, and their median area there is 1936 against 316162 for the fitted blob rules, so they are stalled or tethered (`data/b4_stats.txt`). If fast growers are missing from the census it is for another reason; the only candidates are the 10 grid-escapers among the builders (0.3% of the census). The census contains 52 rules that are exact repeats of a shorter rule (u^k, which behave identically to u); we keep them, since removing them moves the blob median width exponent only from 0.3326 to 0.3325.

2.3 The three-factor decomposition

Per window of S steps let B be the number of steps that start on a boundary cell and ΔA the number that enter new ground. Define

$$c = \frac{\Delta A}{B}, \quad \kappa = \frac{B/S}{P/A}, \quad \frac{dA}{dt} = c \cdot \kappa \cdot \frac{P}{A}. \tag{1}$$

The last equality is an *identity* given these definitions. Hence if c and κ drift as t^{γ_c} and t^{γ_κ} and $P \sim A^\beta$, then

$$a = \frac{1 + \gamma_c + \gamma_\kappa}{2 - \beta}, \quad \alpha \approx \frac{a}{2}. \tag{2}$$

Equation 2 is also algebra, so agreement between the measured a and Equation 2 (median error 0.008 over blob rules) checks the fits for consistency only. The empirical content is in the *values* of the three factors: κ is the fraction of time on the boundary relative to a uniform spread over the territory, c the chance a boundary step breaks new ground. If $\gamma_c = \gamma_\kappa = 0$ and $\beta = \frac{1}{2}$, Equation 2 gives $a = \frac{2}{3}$.

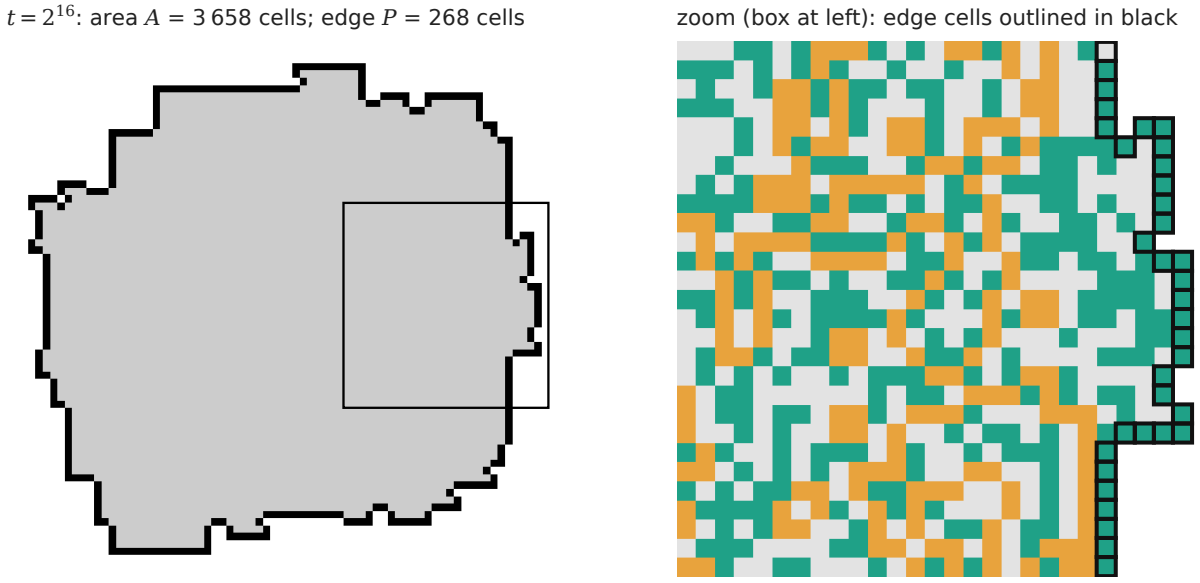


Figure 5: The quantities in Equation 1 on a real territory (RLL, 2^{16} steps). Grey: cells visited so far, area A. Black: the edge, visited cells with at least one never-visited neighbour, size P. Only a step taken from an edge cell can enter new ground; c is the fraction of edge steps that do, and κ compares the share of time the ant spends on the edge with the share of the territory that is edge. The right panel zooms on the boxed stretch of coastline.

2.4 Reproducibility

All simulation is deterministic and seeded (rules start from a blank board, so a rule has one trajectory; seeds enter only in the walkers and the patched-start control). The Rust fast ant used for sweeps is tested step for step against an oracle (`tests/ant.rs`, `tests/probe.rs`). Every number in this paper is computed by `research-outputs/cube-root-growth-paper/compute_numbers.py` from the files in `data/` and written to `research-outputs/cube-root-growth-paper/numbers.json`, which this document reads; figures are made by `research-outputs/cube-root-growth-paper/figures.py`. Bootstrap intervals use a fixed seed. Commands are in `research-outputs/cube-root-growth-paper/README.md`.

3 Calibration against the rotor walker

An exponent fitted over seven octaves from 2^{20} to 2^{26} is not the asymptotic exponent. We therefore ran the identical pipeline on 64 rotor-walker seeds (random initial arrows, 2^{27} steps), whose width exponent is $\frac{1}{3}$ and area exponent $\frac{2}{3}$ [1], and on 16 random walks as a negative control.

	width α	R_g exp.	area a	β
truth (rotor)	1/3	1/3	2/3	–
rotor, median	0.326	0.339	0.687	0.523
rotor, between-seed sd	0.013	0.006	0.008	0.027
ants (core), median	0.333	0.336	0.679	0.525
ants (core), between-rule sd	0.014	0.011	0.022	0.023
random walk, median	0.498	0.467	0.93	0.93

Table 1: Same pipeline on three systems. The rotor has known exponents; the random walk is a negative control (area exponent near 1, boundary exponent near 1). The "ants (core)" rows are selected on $0.30 < \alpha < 0.37$, which truncates their sd: it understates the between-rule spread and flatters the comparison with the rotor's sd. Figure 8, over all blob rules, is the fair comparison.

Findings.

- (i) The pipeline recovers the rotor: the radius-of-gyration exponent is 0.339 (sd 0.006) against the true 0.333, the width exponent is slightly low (0.326) and noisier, and the area exponent overshoots by 0.02. We therefore treat the finite-window bias of a as $+0.02$ and prefer the R_g estimator, which is the least noisy.
- (ii) The random walk is clearly separated (area exponent 0.93).
- (iii) The ants' R_g exponent 0.336 and area exponent 0.68 sit within the rotor's bias and sd.
- (iv) The boundary exponent of both is about 0.523 (rotor) and 0.525 (ants), not 1/2; it is a finite-window effect shared with the rotor and not evidence of a rough edge.

To check the pipeline further we reproduced the published rotor measurements of Kapri and Dhar [8] from 1000 seeds (radius exponent local slopes 0.263, 0.315, 0.306, 0.346, 0.332, 0.334, tending to $\frac{1}{3}$; boundary-width exponent local slopes falling from 0.19 to 0.132, against their reported value $\alpha/3 \approx 0.13$).

A reservation: for the rotor, the true limit exponent is known, but only by proof for the radius; our statements about its β and c are measurements, not theorems.

4 Results: the census

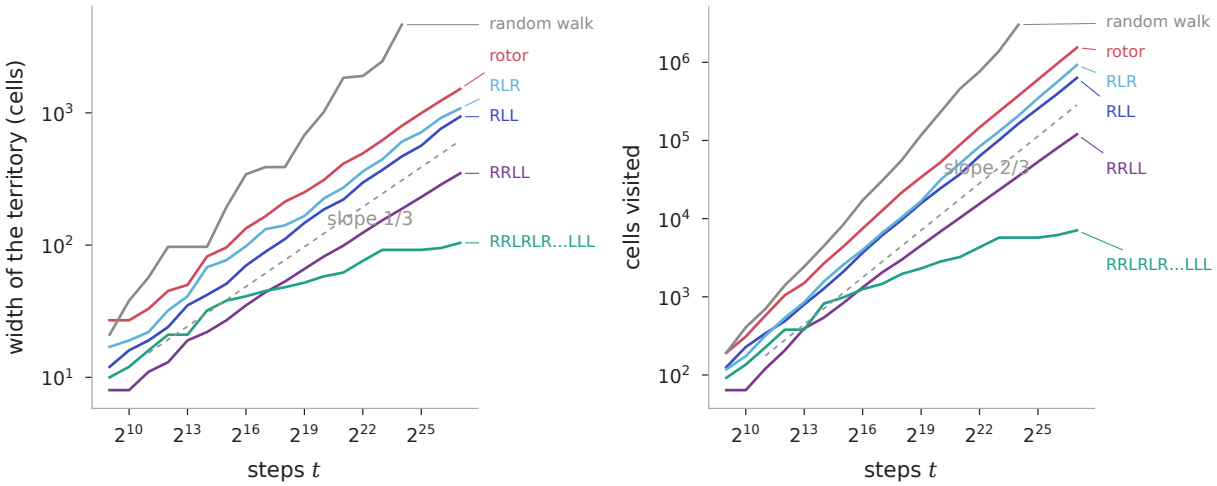


Figure 6: How the territory grows with the number of steps t (log scales, so a power law is a straight line). Three ant rules (blue and purple) and the rotor walker (red) climb parallel to the dashed guide lines, slope $\frac{1}{3}$ for the width and $\frac{2}{3}$ for the visited area; doubling the width takes about eight times as many steps. The log-growing rule RRLRLR...LLL (teal) falls away, and the random walk (grey), the negative control, grows much faster, with width $\sim t^{\frac{1}{2}}$.

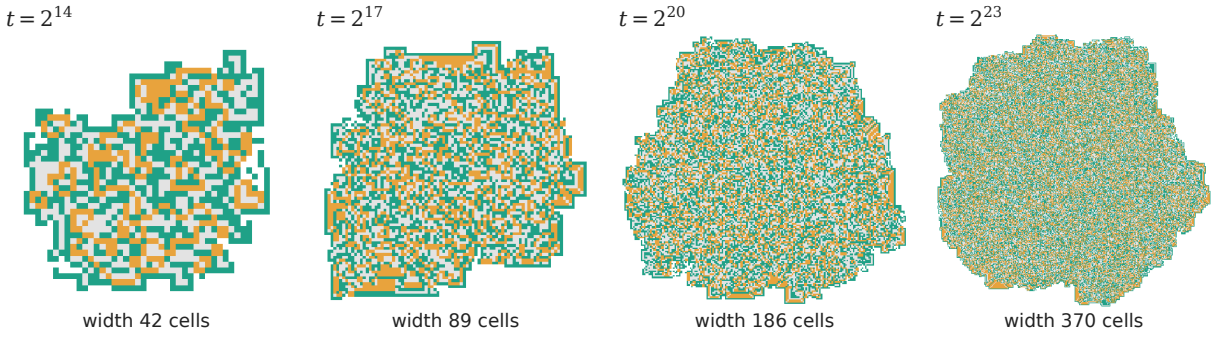


Figure 7: One rule (RLL) at four times, each panel rescaled to the same size. Eight times as many steps buys a territory only twice as wide ($8^{\frac{1}{3}} = 2$), and the shape stays the same round blob: it looks self-similar, and the roundness statistics of Section 7 (anisotropy 0.05–0.07) agree, but we did not test scaling beyond that.

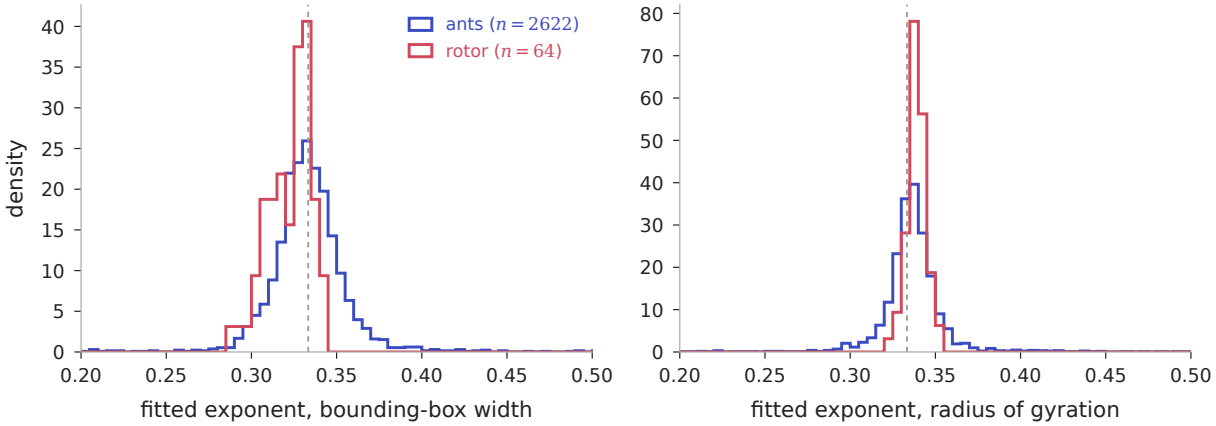


Figure 8: Fitted exponents of 2622 blob-class ant rules and 64 rotor seeds. Dashed line: $\frac{1}{3}$. The ants' distribution is wider than the rotor's: sd ratio 2 (width), 4.3 (R_g).

Exponent. Over the 2622 blob rules the median width exponent is 0.333 (rule bootstrap 95% interval 0.332 to 0.334), the R_g exponent 0.336, and the area exponent 0.679 in the core.

The interquartile range of the width exponent is [0.32, 0.34]. The bootstrap resamples *rules*; it does not include the finite-window bias, which Table 1 puts at about $+0.02$ for a and negative for α.

Against the rotor in the same window the differences (ants minus rotor, medians over all blob rules) are 0.007 for the width exponent, −0.003 for the R_g exponent and −0.008 for the area exponent.

Subtracting the rotor's own bias from the ants' medians gives 0.34 (width), 0.33 (R_g) and 0.658 (area), against $\frac{1}{3}$, $\frac{1}{3}$ and $\frac{2}{3}$. The corrections point in different directions, so the data are consistent with $\frac{1}{3}$ at roughly the ± 0.01 level, not "equal to $\frac{1}{3}$ to three digits"; the correction assumes the ants share the rotor's finite-window bias, which we cannot check.

Rule length. The median is flat in n: Spearman ρ between α and n is −0.018 over blob rules, and per-length medians for $n = 7 \ldots 12$ lie within about 0.01 of one another (table in `research-outputs/cube-root-growth-paper/numbers.json`, `by_n`). The spread between rules does not shrink with n in our sample, so we do not claim a rule-length trend or its absence at larger n.

Long runs. Six typical rules were run to 2^{32} steps; their width exponents fitted from 2^{17} are between 0.324 and 0.34, and neither factor in the decomposition drifts (Figure 9).

Noise. The between-rule scatter is mostly run-to-run: re-running 40 rules from 474 random 5×5 starting patches gives within-rule sd 0.016 for α and a between-rule true sd of only 0.004 (intraclass correlation 0.07).

Genuine rule-to-rule differences in α are small; for R_g and a they are detectable (ICC 0.21, 0.33). We cannot say whether they reflect different exponents or different rates of convergence.

Where the rule-specific part of a sits can be partly located (`data/h1_lacunarity.txt`, `data/h3_collapse.txt`; the 99 random rules of the ICC above, window $k = 20$–26). The area is the product of the two box sides and a filling fraction, so $a = \alpha_x + \alpha_y + \gamma_\eta$ exactly, with γ_η the log-slope of $A/(W_x W_y)$. The rule-to-rule variance of a is carried 43% by α_x, 40% by α_y and 17% by γ_η (covariances of rule means, which include run noise); the two sides rise or fall together (correlation 0.6), while γ_η has ICC 0.11, like the sides taken singly (0.09 and 0.12 a itself 0.34). The territories consolidate slowly ($\gamma_\eta = +0.006$, positive for 85% of rules), but rules do not differ in a mainly because they consolidate at different rates. We have not explained why the sides co-vary by rule.

5 Results: the three factors

For the 2404 core rules (those with $0.30 < \alpha < 0.37$), the medians (and quartile ranges) of the factors are $c = 0.43$ [0.34, 0.48], $\kappa = 0.9$ [0.73, 1.07] and $\beta = 0.525$ [0.511, 0.54], with territory fill (area over bounding box) 0.76 [0.72, 0.82].

The drift exponents are small, but the 0.05 threshold is loose: through Equation 2 a drift of 0.05 shifts a by about 0.033 and α by about 0.017, as much as the rule-to-rule differences we are trying to explain. These fractions therefore say that c and κ drift by less than the effect size of interest, not that they are exactly constant, and only the six 2^{32} runs probe much smaller drifts. $|\gamma_c| < 0.05$ for 100% and $|\gamma_\kappa| < 0.05$ for 83% of all 2622 blob rules (the fractions are computed on the whole class, because "core" is itself defined by a width exponent near 1/3 and would make small drifts nearly automatic).

Figure 9 shows the long runs: both factors plateau from 2^{17} to 2^{32} for six rules. These are the observations that the cube root rests on; the predictions P1–P4 were written down before the analysis (`notes/cube-root-hypothesis.md`, dated 2026-10-10, three days before this draft) and could have failed; P4 predicted β close to $\frac{1}{2}$, and the measured β is 0.525 (core median), equal to the rotor's 0.523 in the same window (Table 1).

The medians quoted here are for the core class, which is selected on the outcome, and are descriptive only.

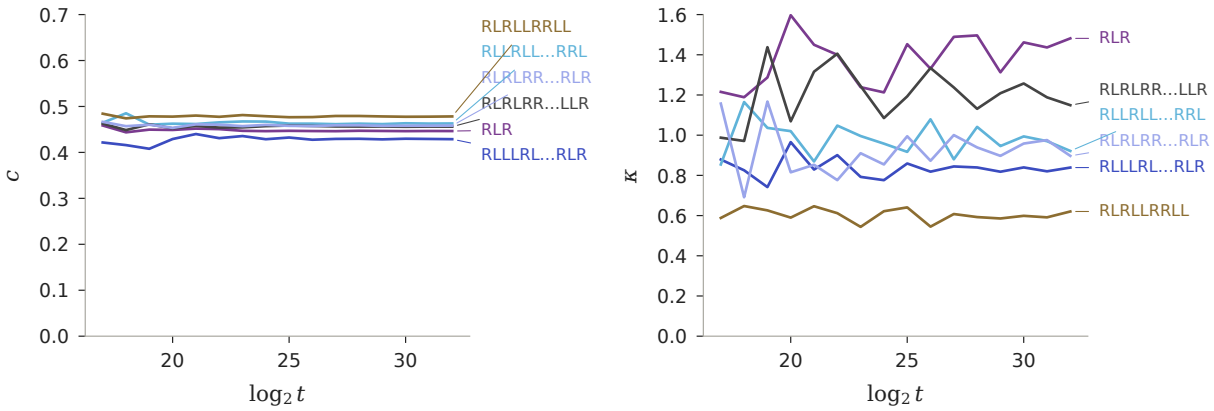


Figure 9: Edge-entry probability c and edge-occupancy ratio κ against $\log_2 t$ for six rules run to 2^{32}. Both are flat after 2^{20}; κ fluctuates by roughly ten percent from octave to octave.

Comparison with the rotor walker. For the rotor the same decomposition gives $c = 0.5$ and $\kappa = 2$; a random walk gives $c = 0.44$. So c alone does not discriminate (the random walk's c is close to the ants'), and the ants' κ (about 0.9) is about half the rotor's.

The cube root does not require the same c and κ as the rotor; it requires that they be constant and that the edge scale as $A^{\frac{1}{2}}$. The ant's edge occupation is *lower* than uniform ($\kappa < 1$ for most rules); we have no explanation of the value of κ.

Is κ a property of the rule? The claim that κ differs between rules moderately reproducibly, yet is not carried by the letters, rests on three measurements, drawn from four data files; the between-rule fractions are intraclass correlations as in the Noise paragraph.

- (i) *Moderately reproducible* (`data/a1b_icc.txt`; the within-rule sd is nearly as large as the between-rule sd): 99 random rules, each run from 12 random 5×5 patches, have a between-rule sd of κ of 0.44 against a within-rule sd of 0.37 in the core band (ICC 0.58; 0.63 for $\log \kappa$; 58 further core rules give 0.50 and 0.63). The width exponent on this design (`data/b2_probe.csv`) has ICC 0.12 (0.065 in the Noise paragraph, a different set of rules and starts), so κ is much more rule-specific than the exponent it controls.
- (ii) *Not carried by the letters* (`data/a1_kappa_proxy.txt`, `data/a1c_predict.txt`): a linear regression on string features (turn probability q, n, run count, letters 1–6), fitted on a random half of the 2622 non-bursty blob rules and scored on the other half, has held-out R^2 = 0.003 for κ and 0.002 for its drift γ_κ, against 0.78 for c with the same features and split (a different, simpler model than the chirality model of the next section, which reaches 0.66 on a larger held-out set); Spearman(κ, q) = 0.08. On 192 rules with at least six core-band runs, leave-one-rule-out R^2 for the rule-mean $\log \kappa$ is -0.06 to -0.04 for ridge on letter-by-position, about 0 for ridge on 2- and 3-letter substring counts, and 0.04 for a prefix-5 neighbour mean.
- (iii) *Not carried by the edge colour mix* (`data/a1_edge_q.txt`): on 40 random core rules the edge-weighted turn probability correlates with c at Spearman -0.84 (a positive control that the instrument can see rule dependence) but with κ at only -0.27, and leave-one-out R^2 of κ on it and the edge-colour distance from uniform is -0.05.

So κ behaves as an emergent property of each rule's dynamics.

Does early transport predict later growth? A same-window identity cannot test persistence. We therefore specified a forecast on existing replicated runs before computing its errors (`notes/transport-forecast-2026-10-13.md`). Write $g = \frac{P}{\sqrt{A}}$; formally Equation 1 gives $\frac{d\left(A^{\frac{3}{2}}\right)}{dt} = \left(\frac{3}{2}\right)c\kappa g$. For an octave ending at t, with duration $S = \frac{t}{2}$, the exact discrete budget is

$$Q = \frac{A^{\frac{3}{2}} - (A - \Delta A)^{\frac{3}{2}}}{S}. \tag{3}$$

The endpoint product $\left(\frac{3}{2}\right)c\kappa g = \left(\frac{3}{2}\right)\left(\Delta\frac{A}{S}\right)\sqrt{A}$ is a right-end approximation, typically 1.112 times the exact budget in this cohort. We forecast with Equation 3. Each trajectory's mean budget over windows ending at 2^{20}, 2^{21} and 2^{22} predicts its increments in $A^{\frac{3}{2}}$ over the next four octaves, ending at 2^{23}–2^{26}, without updating from those outcomes. A competing model extrapolates a log–log drift fitted to the same three early budgets. The rule-independent baseline is the median early rule-mean budget after excluding the target rule and all its seeds.

There are 1185 complete trajectories across 99 rules, with approximately 12 patched starts per rule; 15 trajectories fail the completeness, nonescape or positive-growth criteria. We apply no outcome exponent filter. For each trajectory, error is the median absolute relative increment error over four forecast windows; the summary is the median of equal-weight rule means. Constant budgets give 8.5% error against 26.9% for the baseline and 23.3% for the drift fit. The reduction is 68.5% (rule-bootstrap 95% interval 61.3–80.6%; 2000 resamples, seed 20261013), passing the stated 30% prediction. Integrating all four forecast increments from the observed area at 2^{22} gives final-area error 5.4% against 17.6% for the baseline.

Calibration separates persistence from a universal forecasting advantage. The same estimator on 64 rotor seeds, with matched windows, gives 6.1% error against 5.7% for the other-seed baseline. On 16 random-walk seeds (training $k = 17$–19, forecast $k = 20$–23), errors are 60.3% and 58.8%; drift extrapolation still misses by 47.9%. Synthetic $A = Ct^{\frac{2}{3}}$ and $A = Ct$ recover their known constant and square-root budgets respectively.

This typical accuracy hides severe failures. RMS within-rule seed sd of relative increment errors is 188.68 for constant budgets, 918.8 for drift and 8.67 for the baseline; these are ratios, not percentages. One retained trajectory (`RLLRRRLLRLRR@108`) has constant-budget error about 6506 times the observed increment. Near-stalls make relative errors enormous, so the robust improvement does not imply uniformly reliable predictions. Escaped or incomplete trajectories are excluded, and this test cannot forecast their breakouts. An exploratory check, specified after the primary result, uses the median early budget of the other seeds of the same rule and gives 7.4% error, slightly better than each trajectory's own early budget. This supports persistent rule-level transport in the measured windows; it neither identifies the spatial-memory mechanism nor proves asymptotic scaling. Exact recipes, input hashes and all trajectory errors are in `data/transport_forecast.py` and `data/transport_forecast.json`.

A memoryless control. We then tested the mechanistic candidate directly: a memoryless walker that turns right with probability Q and left otherwise, independent of the board, driven through the same probe (`data/a1d_stats.txt`; $Q \in \{0.25, 0.4, 0.5, 0.6, 0.75\}$, six seeds each, 2^{26} steps on a 32768^2 board; predictions written beforehand in `notes/frontier-2026-10-13-results.md`, A1d). Its κ is roughly constant at 1.3 to 1.45 (within $\pm 5\%$ across Q from 0.25 to 0.6), but its area exponent is 0.94 to 0.96, not $\frac{2}{3}$.

So independent persistent turns reproduce roughly uniform occupancy but neither the cube root nor the ants' κ (about 0.9) nor the rotor's (about 2.0). This rejects the independent-turn mechanism; it motivates testing board memory without establishing that memory alone is sufficient, or that every memoryless mechanism must fail.

This rules out one explanation of κ; it does not supply another.

What does stochastic board memory change? Figure 10 makes three comparisons concrete. A once-excited walk uses a directional cookie on its first departure from each site, then moves uniformly on revisits [9]. Randomly orienting those cookies removes the ensemble drift. A linearly edge-reinforced walk instead favours edges it has crossed: each undirected edge starts at weight 1 and gains I on each crossing. Neighbour choices are proportional to those weights [10]. These models separate one-use site memory from persistent edge memory; the random-arrow cookie is our specified variant of the directional model.

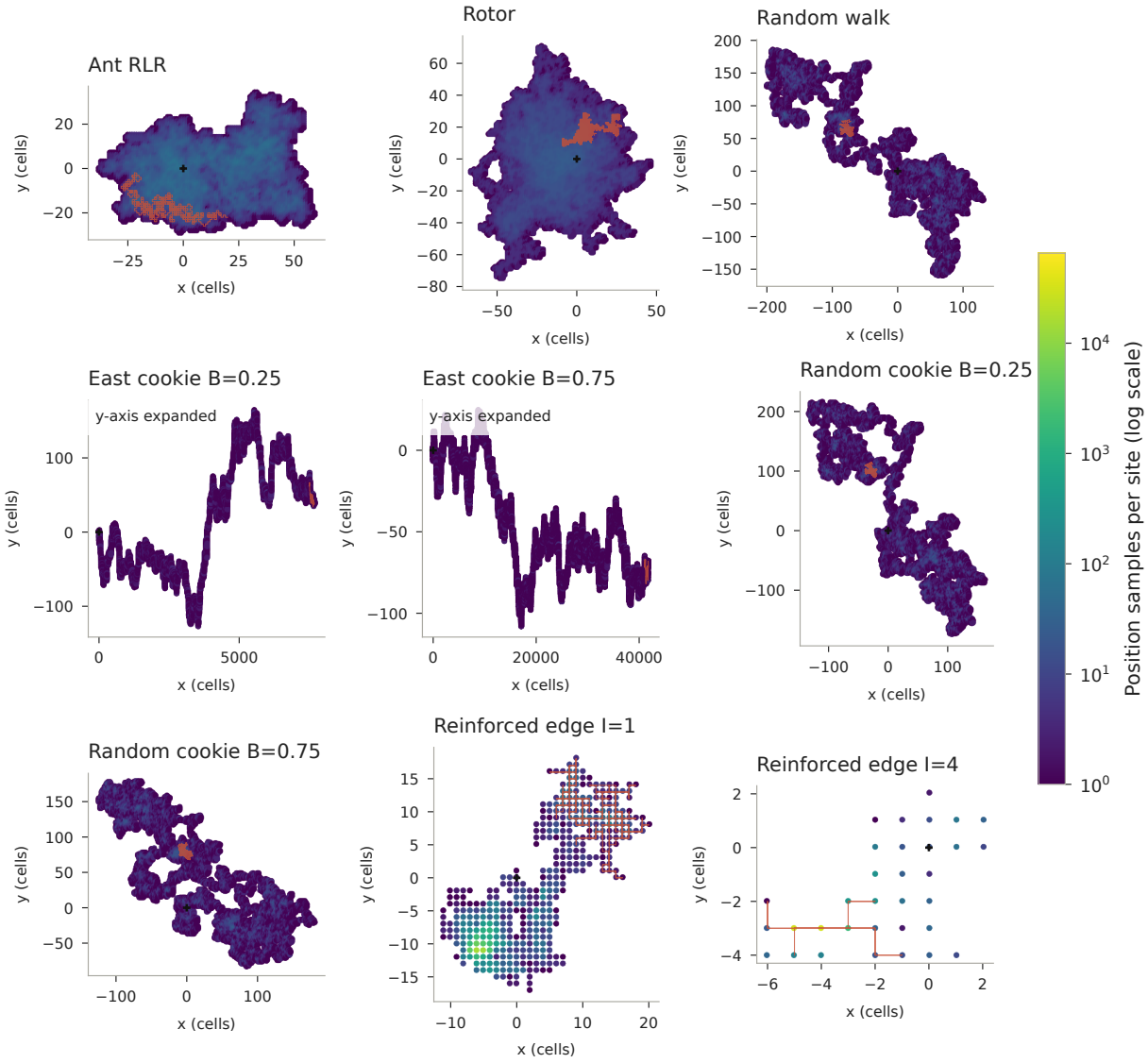


Figure 10: Nine walkers after the same 65,536 steps. Stochastic walkers use the preselected seed 301; the ant starts blank. Colour records position samples per site (including the origin); orange traces the final 1,024 steps and + marks the origin. Panels have equal x/y scaling except the directional-cookie panels, whose y-axis is expanded to show transverse motion. Spatial scales differ: read the cell coordinates. Directional cookies spread east, random cookies resemble the random walk, and strong reinforcement repeatedly uses a small territory. These single trajectories illustrate behaviour, not the ensemble exponents. B is the first-departure probability of following the cookie instead of choosing a uniform neighbour; I is the edge-weight increment.

The registered pilot used eight seeds (301–308) per setting, up to 2^{20} steps on an 8192^2 board. Table Table 2 reports per-seed fits over $k = 14$–18 and κ at 2^{18}; uncertainties are between-seed sd. All 16 directional-cookie runs escaped before completing that window, so we report no fitted exponent for them. Random-arrow cookies stayed close to the random-walk control. Weak edge reinforcement had a mean area exponent near $\frac{2}{3}$, but its large scatter and falling κ do not establish the ant regime. Strong reinforcement greatly reduced exploration.

Walker	Area slope a	κ at 2^{18}
Random walk	0.924 ± 0.043	1.350 ± 0.082
Rotor	0.740 ± 0.027	1.845 ± 0.118
Random cookie, $B = 0.25$	0.942 ± 0.047	1.327 ± 0.075
Random cookie, $B = 0.75$	0.920 ± 0.044	1.293 ± 0.094
Edge reinforcement, $I = 1$	0.698 ± 0.303	0.312 ± 0.132
Edge reinforcement, $I = 4$	0.187 ± 0.092	0.328 ± 0.447

Table 2: Stochastic memory pilot, eight complete seeds per row. The early rotor area slope overshoots $\frac{2}{3}$ by 0.074, so this window cannot identify an asymptotic exponent. In the later $k = 16$–20 window, the rotor gives 0.715 ± 0.019, weak edge reinforcement 0.701 ± 0.277, and strong reinforcement 0.216 ± 0.096. Weak reinforcement's mean κ falls to 0.155 ± 0.084 at 2^{20}.

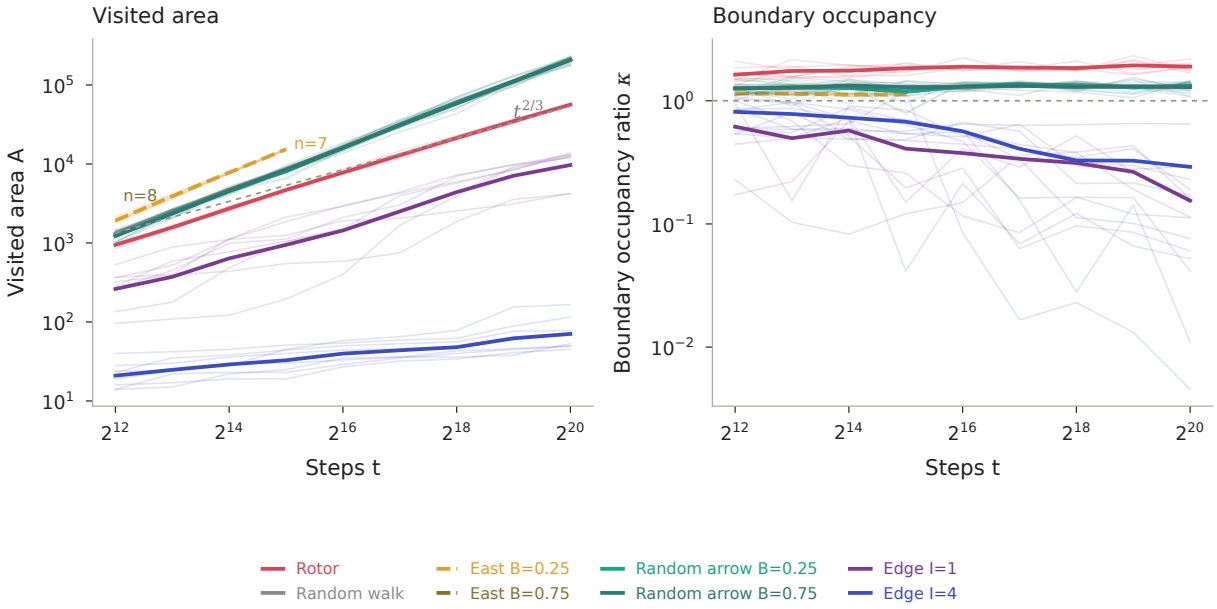


Figure 11: Growth and boundary occupancy across the same pilot. Thin lines are individual seeds; thick lines show the geometric mean area and arithmetic mean κ. All solid curves have eight seeds. Dashed directional-cookie curves end when runs escape; terminal labels give the number still contributing, and later points describe survivors only. The grey guides show area proportional to $t^{\frac{2}{3}}$ and $\kappa = 1$. Persistent memory can suppress exploration without producing stable ant-like boundary occupancy.

The prediction that random cookies at $B = 0.75$ would give a between 0.60 and 0.75 failed. Weak reinforcement passed that area band but failed the predicted κ band of 0.5–3. Thus none of these settings reproduced both features of the target regime. This is a finite-window negative result, not a general exclusion of stochastic memory. Definitions, predictions, recipes and all per-seed fits are in `notes/walker-comparisons-2026-10-13.md` and `data/memory_walkers_pilot/summary.json`.

We also measured where the walker stands relative to the edge (`examples/layers.rs`; design and predictions in `notes/kappa-experiment-design.md`, written beforehand). Let layer d be the Manhattan distance to the nearest never-visited cell minus one, so layer 0 is the edge, and let $\rho(d)$ be the visit rate per cell at layer d relative to the territory mean ($\rho(0) = \kappa$ by construction, which we checked: the layer-0 count equals the probe's perimeter in every row). For the rotor (12 seeds, $k = 22…25$) ρ is 1.8 at layer 0, 1.1 at layer 1 and 0.9–1.0 deeper; for the memoryless walker it falls smoothly from 1.4 to 0.8 over four layers. For the ants it does not have a single shape. In 24 ant rules (12 chosen as the lowest and highest rule-mean κ and 12 closest to 0.9, blank board, `data/layers_stats.txt`, `data/layers_stats_typical.txt`, `data/layers_classes.txt`) we find two classes. In 15, a *crust*: layers 2 to 4 are almost never visited (ρ between

0.01 and 0.6), the walker concentrates in layers 0 and 1, and for 13 of them c is 0.61–0.66 and the mean number of edge-phase visits per cell is 1.5 to 1.6 (the other two have c 0.44 and 0.48), yet κ ranges from 0.13 to 18. In 9, the occupation is uniform to within 30% at layers 2 to 4, c is 0.27–0.48, and κ lies between 0.82 and 0.96 (a range truncated by how those rules were chosen). A proxy for the crust class in the census, $c \geq 0.6$, picks out 12% of blob rules (319 of 2622, including a spike of 263 with c between 0.65 and 0.70) whose median κ (0.80) is ordinary. Two predictions made beforehand failed: that the κ difference between rules lives in the first two layers (it did not, for 15 of 24), and that κ follows the edge dwell time (Spearman -0.26 for the 12 extreme rules, -0.59 for the 12 typical, the wrong sign). The two-class split was found after looking and is exploratory. Its consequence is that κ is not fixed by the edge-entry statistics: within the crust class c and the dwell are constant while κ spans more than an order of magnitude. This characterises the unexplained quantity; it does not explain it. A second intervention gave a result that we did not predict. A rotor whose arrow is replaced by a random one with probability ε per visit (`data/armb_stats.txt`; ε from 0 to 0.1, eight seeds each, 38 of 48 runs fitted, $k = 20\ldots26$) reproduces the rotor at $\varepsilon = 0$ (area exponent 0.69, $\kappa = 2.0$), but already at $\varepsilon = 0.001$ the area exponent is 0.76 and by $\varepsilon = 0.03$ it is 0.94, earlier than we predicted. κ did not fall towards the memoryless value 1.4 as predicted; it rose to 3.6, 5.8 and 6.7 for $\varepsilon = 0.001, 0.003, 0.01$ and fell back at larger ε, with a positive drift in time, so these values are transient readings during the crossover. Meanwhile c stayed at 0.498 throughout. Exact deterministic memory is therefore what holds κ at 2 in the rotor, and κ and c vary independently, as in the ants' crust class. A third test used 26 pairs of rules with equal length, equal number of R and edge-entry probability within 0.02 but rule-mean κ differing by at least 0.5 (`data/armc_stats.txt`): the depth of the depleted crust does not separate them (the lower-κ member has the deeper crust in 13 of 26 pairs, against the 18 we required). The blank-board κ orders 25 of the 26 pairs as the patched-start rule means do, so the rule dependence is robust to the kind of run. Five candidate predictors and one intervention have now failed to account for it.

Deviating rules. 425 rules grow logarithmically (α median 0.101), with κ median 0.001 and $\gamma_\kappa = -0.79$: the ant rarely visits the edge and the occupancy of the edge decays, consistent with an ant confined near its origin.

103 blob rules have $\alpha > 0.37$ with γ_κ median 0.17 (edge occupation rises). Across blob rules the correlation of α with γ_κ is 0.78, with γ_c 0 and with β 0.2 (Figure 12).

So the outliers are outliers because the ant's occupation of the edge drifts, which is the failure condition we wrote down in advance for the mechanism ("κ drifting strongly with t"). This is a restatement through Equation 2, not an explanation: *why* κ drifts for these rules is unexplained here.

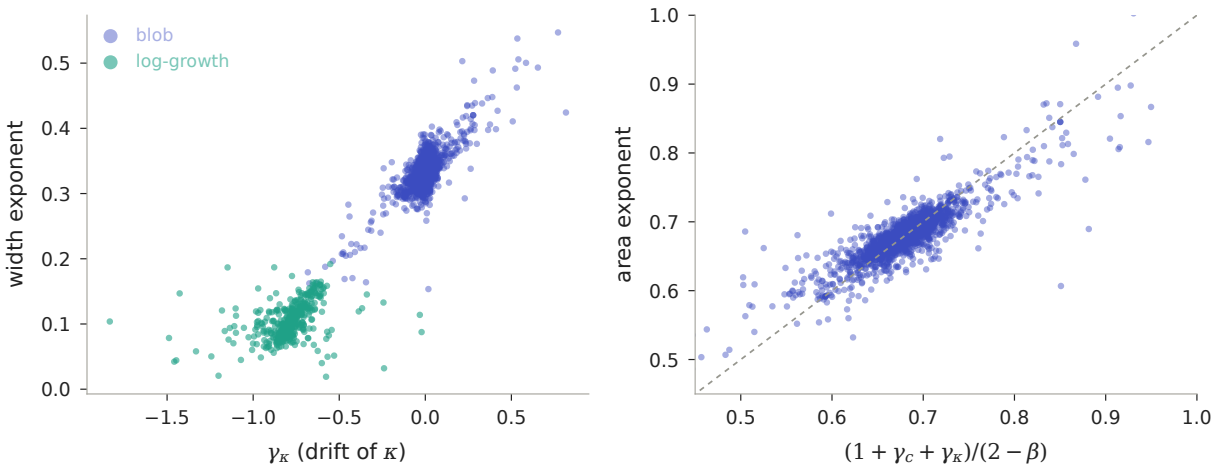


Figure 12: Left: width exponent against the drift exponent of κ. Right: area exponent against the value predicted by Equation 2; the agreement is by construction up to fit error (see text).

6 Results: what sets c, and is the ant well mixed?

Edge-entry probability. Across rules c ranges from about 0.1 to 0.67. A chirality model predicts it from the rule's letters (this is the model behind the abstract's figure; the 0.78 quoted in the previous section is a plain linear regression on string features, with different features, sample and split, so the two R^2 values are not comparable): the edge is chiral (for edge steps on colours 1 to 3, the share with fresh ground on the left exceeds that on the right in 294 of 300 calibration rules; medians 0.54 against 0.17), and so a rule that turns L on a young colour enters new ground more often.

Constants (the share of edge steps by colour and the left/right fresh-ground rates) are fitted on a 300-rule sample and applied to the others without refitting.

On 2747 held-out rules the model attains $R^2 = 0.66$ (correlation 0.87 median absolute error 0.04); grouping rules by their first letter after the leading R explains 0.59 of the variance of c, by the next three letters jointly 0.81, and by the sixth letter alone (rules of length at least 8) 0.001.

The model was a post-hoc fit to these data, though evaluated on rules outside its calibration sample; we have not tested it on rules of length above 12.

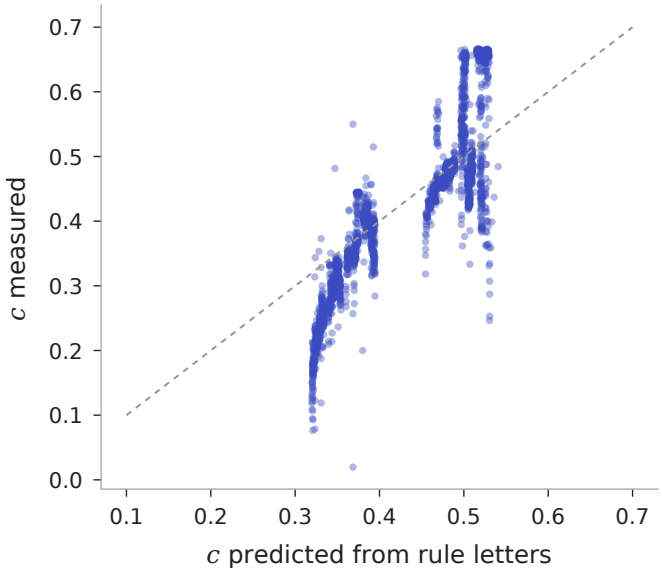


Figure 13: Held-out edge-entry probability: model against measurement. The model takes few distinct values; the R^2 is mostly between-cluster, and within the clusters near 0.33 and 0.53 it discriminates poorly (rules with predicted $c \approx 0.53$ have measured c from about 0.25 to 0.67).

Mixing. The mean-field step assumes the ant is spread roughly evenly over its territory, behaving like a persistent random walk with turning probability $q = \#R/n$. The sample is the 49 rules in `data/mix_rules.txt`, of which 33 are blob rules with $0.30 < \alpha < 0.37$; the procedure used to choose the list was not recorded, so it should be read as a convenience sample.

We therefore repeated the test on 40 rules drawn uniformly at random (fixed seed) from the core class, with predictions written beforehand (`notes/mixing-random-sample.md`): 35 of 40 have near-uniform interior colours (our prediction of 60–85% was exceeded), their MSD-slope ratio has median 1.01 (range 0.84 to 1.58), and the remaining 5 have median ratio 1.3.

The convenience-sample numbers follow. For the 33 typical rules we tested (a) colour uniformity (total-variation distance of interior colours from uniform), (b) turn correlations, and (c) the slope of mean-square displacement against lag, which for independent turns is $\frac{1-a'}{1+a'}$ with $a' = (2q-1)^2$.

For 24 rules with near-uniform interior colours (TV < 0.05) the measured slope is within [0.95, 1.54] of the prediction (median 1.04), the displacement exponent is 1, and the lag-1 turn correlation is −0.002 at the median.

For the other 9 rules, whose interior colours are far from uniform, the MSD slope is off by a factor of 1.32 at the median (correlation between the ratio and the colour deviation across the sample: 0.69).

Time to mix across the territory is a fraction 0.002 (median) of the growth time.

The ant is therefore well described as a random walker in a random environment for 73% of the sampled rules and not for the remainder. That the ant is spread out in time does not by itself imply that its edge occupation is κ; it only makes κ a stationary quantity, and Figure 9 shows that.

7 Long ensembles, the occupation profile and the outliers

Three further measurements, produced by separate scripts (`data/item4_stats.py`, `data/shape_stats.py`, `data/item21_stats.py`; hypotheses and predictions are in `notes/frontier-results.md`, written before each run), bear on the weakest points above.

Is the exponent universal? Eleven rules (stratified by length 3–12) were run from 251 random 5×5 starting patches, 24 seeds each, to 2^{30} steps. The local area exponent, averaged over seeds and over the last six three-octave windows, has mean 0.683 across rules (sd 0.014, range 0.66 to 0.709).

The same estimator applied to 24 rotor seeds, whose exponent is proved to be $\frac{2}{3}$, gives 0.678: even the rotor sits 0.01 above $\frac{2}{3}$ at 2^{30} by this estimator. 6 of the 11 rules are within 0.006 of the rotor; the others are up to 0.02 above it or 0.017 below it by amounts many standard errors wide.

One rule (`RRLLLLLLR`) does not self-average: its seed-to-seed spread rises with time. So the data are compatible with a common exponent $\frac{2}{3}$ with slow, rule-dependent convergence, but they cannot distinguish that from rule-dependent exponents within about 0.02.

The occupation profile. The time-averaged visit density, as a function of distance from the centroid in units of the territory radius $R_{\text{eq}} = \sqrt{A/\pi}$ and of t/A, is a cone, $n(r) = 3\big(1 - r/R_{\text{eq}}\big)t/A$, for ants and rotor alike. Over the first ten of fifteen radial shells the ants' profile (88 runs of 11 rules, 2^{28} steps) and the rotor's (40 seeds, 2^{26} steps) differ by at most 5%, and the ant profile differs from the cone by at most 5%; the across-rule relative sd at mid-radius is 4%. This is a prediction we had expected to fail (we had written that the ant's profile would differ from the rotor's). The agreement is nonetheless not independent evidence for the cube root, because the cone is implied by it.

Suppose the walker is spread uniformly over the current territory, $n(r, t') = 1/A(t')$ for $r < R(t')$, and $A(t) \propto t^a$. A cell at radius r is inside from the time t_r at which $R(t_r) = r$, so its accumulated visits are $\rho(r) = \int_{t_r}^{t} dt'/A(t') = (t/((1-a)A))\big(1 - (r/R)^{2(1-a)/a}\big)$. For $a = \frac{2}{3}$ this is exactly $(3t/A)(1 - r/R)$, and it integrates to t over the disc. The cone is therefore the footprint of a uniformly mixing walker inside a disc growing as $t^{\frac{2}{3}}$; nothing specific to rotors or ants enters beyond those two assumptions (`notes/frontier-2026-10-13-results.md`, A3). It also explains why ants and rotor agree although their κ differ (about 0.9 against 2.0): κ changes only a thin edge layer, not the bulk integral. What the profile adds is a check of uniform occupancy in the bulk.

With the measured rotor exponent $a = 0.687$ the formula gives a centre prefactor $1/(1-a) = 3.19$ rather than 3, against a measured 2.87 (`data/shape_stats.txt`) that the 0.075 R_{eq} shell width lowers by roughly 3–4%, so the data do not separate $a = \frac{2}{3}$ from 0.687 at better than about 5%. The formula's predictions for other exponents were tested on walkers with $a \neq \frac{2}{3}$ (`notes/frontier-2026-10-13-results.md`, W2, W3 and the checks after them; `data/w3_stats.txt`, `data/w_checks.txt`), with results that do not support the derivation beyond $a = \frac{2}{3}$. (i) Rotors whose arrow is randomised with probability $\varepsilon = 0.001$ ($a = 0.75$–0.78, four seeds, territories round enough for a radial profile: anisotropy ≤ 0.31, fill 0.96) have a centre density of 1.9–2.1 against 3.5–3.8 from the formula, so the formula fails and the cone, at 3, is also off. For $\varepsilon \geq 0.003$ and for a memoryless persistent walker the territories are elongated or holey (anisotropy up to 0.9, fill down to 0.3), so the premise fails first and the profile cannot test the formula. (ii) Two chaotic 2-state turmites with a stable $a \approx 0.74$ over six octaves (`tm:2:2:46382`, `7982`; single trajectories) keep a cone-like centre (2.8 and 3.1) below the formula's 3.5 and 3.2. So the cone is

not simply a consequence of "uniform occupancy and an exponent", and we cannot say that uniform occupancy needs $a = \frac{2}{3}$; every walker we tried has $a \geq \frac{2}{3}$, and the exponents are finite-window values. We had also meant to test the formula on the $a \approx \frac{1}{2}$ turmite cluster, but those tables grow as one-dimensional lines (anisotropy 1.00), and eight of the ten sampled are the same trajectory, so the cluster does not bear on the question.

The territories are round (median anisotropy 0.07 at 2^{28}, the rotor's 0.054 at 2^{26}; fill 0.98).

The surface width does *not* match: its local exponent fitted over $k \geq 18$ is 0.26 for ants and 0.16 for the rotor (Kapri and Dhar: about 0.13), so the ants' edge is broader at equal size. These single fits use a different window ($k \geq 18$) from the matched-k follow-up below, which is why the absolute values differ; the comparison that matters is the ratio.

A longer follow-up (`data/a4c_stats.txt`; 36 ant runs and 14 rotor seeds to 2^{34}) finds both exponents flat rather than falling, about 0.22–0.25 for ants and 0.10–0.13 for the rotor at every matched k from 22 to 32, so the gap of roughly a factor two does not close in the accessible range; whether it closes asymptotically is undecided. The gap is not an artefact of the extreme-κ rules with a "crust" (an almost unvisited shell between the active edge and the bulk, found in 15 of 24 rules examined; see Section 5): to 2^{30} the 9 ordinary-κ rules without a crust have local exponents 0.21–0.35 (median 0.27), all above 0.18, against 0.17–0.45 (mean 0.32) for the 15 crust rules (difference +0.04, Mann–Whitney $p = 0.054$; Spearman with $\log \kappa$ +0.40, $p = 0.056$; one trajectory per rule, W1). The wider ant edge is thus generic to ants, with at most a mild extra roughening in the high-κ tail.

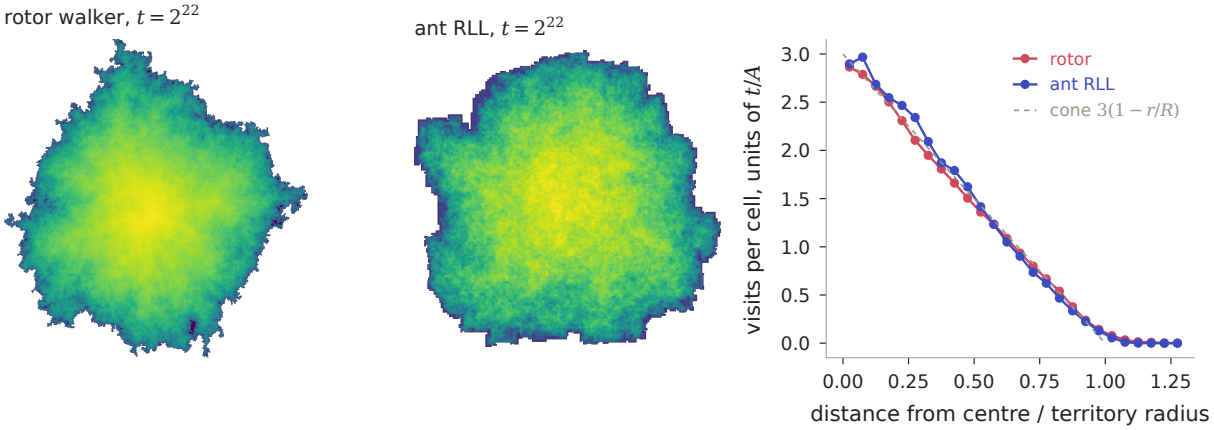


Figure 14: Left and middle: how often each cell was visited (log scale) in a single run of 2^{22} steps, for the rotor walker (one seed) and the ant RLL. Right: visits per cell, averaged in rings around the centre, against the cone $3(1 - r/R)$. Both curves follow the cone; this figure is one run each and is an illustration, not the ensemble measurement quoted in the text.

Width against radius. Refitting the same A4/A4c trajectories directly against $R = \sqrt{A/\pi}$ gives $W_{\text{surf}} \propto R^{\zeta}$ with rotor $\zeta = 0.401$ (14 seeds, sd 0.073; bootstrap mean interval 0.366–0.437) and ant $\zeta = 0.714$ (36 runs across six rules, RMS within-rule seed sd 0.134; rule-bootstrap interval 0.664–0.761), over $k = 22$–32. The rotor prediction $\zeta \in [0.30, 0.37]$ failed: this estimate is closer to $\frac{3}{8}$, implied by $W \propto t^{\frac{1}{8}}$ and $R \propto t^{\frac{1}{3}}$, than to $\frac{1}{3}$. Restricting to $k = 26$–32 gives 0.389 and 0.685; the later A4c cohort alone over $k = 26$–34 gives 0.395 and 0.637. The ant/rotor slope ratio therefore changes from 1.78 to 1.61. Replacing R with R_g or the mean outer radius changes these means by less than 0.005. This confirms a finite-window gap but does not support an exact factor-two law. These are growth-trajectory slopes, not spatial roughness exponents; height correlations and removal of global shape modes are needed to test a KPZ or other interface universality class (`data/width_radius_refit.json`).

Are the outliers noise? We re-ran the 36 fastest and 32 slowest blob rules from 12 random patches each. If their exponents had been run-to-run noise they would regress to the middle. They mostly do not: only 17% of the fast rules fall below 0.37 and 34% of the slow rules rise above 0.30; the Spearman correlation between census and re-measured exponent is 0.84, and the re-measured exponent correlates

with γ_κ at 0.82. So the outliers are real properties of the rules over 2^{20}–2^{27} steps, with edge occupation drifting; whether they are different asymptotes or very slow transients is untestable at this length (the previous paragraph shows even typical rules differ by about 0.02 at 2^{30}).

This revises our earlier statement that most between-rule scatter is noise: for typical rules it is (within-rule sd 0.016 against a between-rule sd of 0.004), but the tails are not.

8 What is tested and what is not

Claim	Status	Evidence
Width $\sim t^{\frac{1}{3}}$, area $\sim t^{\frac{2}{3}}$ for non-highway rules	Measured; consistent after the rotor-calibrated bias	Table 1, Figure 8
c constant in time	Tested, not refuted	Figure 9, γ_c
κ constant for typical rules	Tested, not refuted (drift below 0.05) for 83% of blob rules; refuted for log and fast classes	Figure 9, Figure 12
$P \sim A^{\frac{1}{2}}$	Not confirmed: measured $\beta = 0.525$, equal to the rotor's under the same window	Table 1
Equation 2 predicts a for outliers	Algebraic identity; not a test of mechanism	Figure 12 (right)
Edge-entry probability c from rule letters (chirality model)	Held-out $R^2 = 0.66$; length ≤ 12 only	Figure 13
Well-mixed ant	True for 24 of 33 convenience-sampled rules and 35 of 40 random core rules	Mixing paragraph
Occupation profile is the rotor's cone	Measured; predicted to fail, did not (within 5% of the rotor). Derived as a consequence of uniform occupancy and $A \sim t^{\frac{2}{3}}$, so not independent of the cube root. **Unexplained**: why the profile is the cone. The derivation from uniform occupancy plus the exponent fails for walkers with $a \neq \frac{2}{3}$ (noisy rotor, random-background ants), while two turmites with $a \approx 0.74$ keep a cone-like centre	Section 7
One exponent for all rules	Neither confirmed nor refuted: rules differ by up to 0.02 at 2^{30}, as does the rotor from its proved value	Section 7
Outliers are noise	Refuted for most: 17% / 34% regress	Section 7
Surface width matches the rotor	Not supported: ants' exponent is about twice the rotor's (0.26 against 0.16 for a fit over $k \geq 18$; 0.22–0.25 against 0.10–0.13 at matched k = 22–32)	Section 7
κ is rule-specific but not predicted by the rule string	Moderately reproducible (ICC 0.58); held-out R^2 0.003 (letters) against 0.78 for c	Section 5
Why κ takes its value, why it drifts in some rules	**Unexplained**; a memoryless persistent walker (A1d) gives $\kappa \approx 1.4$ and area exponent 0.95, rejecting that memoryless control; edge-layer anatomy (A1-typical) separates a crust class from a uniform class but does not predict κ (predictions failed); κ is not shared between cyclic rotations of a rule (ICC 0.20, K1); a stochastic true self-avoiding walk gets κ up to 1.97 with area exponent 0.98 (E1), so $\kappa \approx 2$ is not specific to the rotor	Section 5

The prefactor of $A \sim t^{\frac{2}{3}}$ is set by $c\kappa$	Partly: dividing by $c\kappa$ cuts the between-rule variance of $\ln\left(A/t^{\frac{2}{3}}\right)$ at 2^{26} by 59% (pre-registered level 75%); a free fit gives power 0.7, $R^2 = 0.73$. c and κ come from the same runs as A, so part of this is the identity of Equation 2	`data/h3_collapse.txt`
Rule-specific area exponent is filling-fraction drift	Refuted: γ_η explains $R^2 = 0.45$ of $a - 2\alpha_w$ and has ICC 0.11 (predictions ≥ 0.80, ≥ 0.35); the two box sides co-vary instead	`data/h1_lacunarity.txt`
Whether the edge exponent is 1/2 asymptotically	**Unexplained**; the finite-window β is 0.525 for ants and 0.523 for the rotor; to 2^{36} the ants (six rules, two starts each) read 0.509 ± 0.003 against the rotor's 0.501 ± 0.003, both close to 1/2, but the decision rules we set in advance are inconclusive (W4)	

9 Limitations

- **One trajectory per rule.** The ant is deterministic from a blank board; the only noise estimate is the patched-start control (40 rules, 474 runs). Rules are not independent samples of anything; the bootstrap interval is a statement about this set of rules.
- **Finite windows.** All exponents come from seven octaves; the rotor calibration quantifies but does not remove the bias, and the rotor's bias need not equal the ants'. Long runs to 2^{32} cover only six rules.
- **Rule length ≤ 12.** Both the census and the c model stop there. Rules with large n are the majority of all rules; if the cube-root regime is a property of short rules, we cannot tell.
- **Edge exponent at finite time.** A separate check on the six long runs (`data/edge_exponent.py`, output `data/edge_exponent.txt`) gives local boundary exponents of P against A (slope over five octaves from k; a different estimator from the census β, hence 0.51–0.53 rather than 0.525), averaged over the six rules, of 0.53 at $k = 20$, 0.51 at $k = 26$ and 0.52 at $k = 28$, with single rules between 0.48 and 0.55 and no clear fall; the rotor's local slope is 0.52 at $k = 22$. A pure $A^{\frac{1}{2}}$ law fits worse than $P = c_1\sqrt{A} + c_0$ (rms log-residual 0.045–0.16 against 0.018–0.073 over the six rules), so the data are consistent with a square-root edge plus an offset shared by both walkers, but do not establish the asymptotic value. Extending the six ant rules (blank and one patched start each) to 2^{36} gives a mean local slope of 0.509 ± 0.003 (twelve runs; the rotor at 2^{36}: 0.501 ± 0.003), with a fall of 0.009 ± 0.004 over the last two octaves against the rotor's 0.012 ± 0.004; our pre-set rules for "consistent with 1/2" and "plateau above 1/2" both return inconclusive, and the fall again rests on the last octave.
- **Exclusions.** 153 rules were not fitted and 8 flagged bursty; the builder and log classes use thresholds chosen by eye.
- **Highway builders.** Classification is by finite simulation. A rule counted as chaotic may build a highway after 2^{27} steps (Langton's ant itself takes about 10^4). Several hundred rules of length ≤ 10 were left unresolved in an earlier highway census (`data/unresolved_le10.txt`). Of 807 such rules run to 10^{10} steps, 38 (4.7%) became highways in $(10^9, 10^{10}]$. In a 240-rule sample run to 10^{11} only 3 (1.25%) did in $(10^{10}, 10^{11}]$, so the tail is thinning, not flat, but one decade cannot show that the set of late highways is finite (`data/b5_stats.txt`; the other 492 survivors were not run in that sample). A rule that stays on the board is not necessarily in a stable regime either: `RLRLLL`, one of the long-run rules, ran as a blob until step 1.2677×10^{10} and then switched within a few 10^6-step windows to a steady builder at 4.02 steps per newly visited cell (all 299 later windows), and left a 32768^2 board at 1.2974×10^{10} (`examples/ribbon_trace.rs`, `data/g1_stats.txt`; one blank-board trajectory, trigger unknown). Two 4×10^5-step trajectory dumps at 1.28×10^{10} and 1.29×10^{10} (`examples/builder_dump.rs`, `data/g1_builder_*.py`; notes/g1-builder-structure-2026-10-14.md) show that this builder is a boustrophedon column sweeper rather than a perimeter crawl: it makes a slow traverse of the full height L (6,400

rows at the first dump and 8,610 at the second) at about 10 steps per row, a fast return at about 2 steps per row, and adds three columns per round trip, and the wake is exactly periodic in x with period 3 (column agreement 0.99, colours 3, 4, 5 only). Three columns of L cells per $\approx 12L$ steps is 4 steps per new cell, and $dx/dt = -1/(4L)$ with the measured $L \approx 0.67\ |x| - 960$ predicts $d|x|^2\ /dt = 0.86$ and 0.84 at the two dumps against 0.86 and 0.83 measured, so $|x|$ grows as $t^{1/2}$ in this phase. The two values of L come from the same single trajectory, so this is a consistency check on the sweep picture, not a test of the $L \sim |x|$ law; why the pattern has period 3 and what triggers the switch are unexplained. Three further board escapes without a recognised highway in the 10^{11} sample may be of this kind but were not traced.

- **Mechanism is a decomposition, not a derivation.** We do not derive the cube root from the rule; we show it follows from three measured quantities each of which is constant in time for most rules.
- **Related but untested ideas.** Experiments in which the ant's territory was replaced by statistically matched fields (pair, 2×2, 3×3) suggest that the log-growing "tethered" rules depend on the exact configuration of the territory rather than on local statistics. That work was run on two rules at one age with one trajectory each, and a follow-up knockout map has not been run. We draw no conclusion from it in this paper.

10 Conclusion

Non-highway generalised Langton's ants of length 3–12 grow, to within the bias our calibration measures, as $t^{\frac{1}{3}}$ in linear size and $t^{\frac{2}{3}}$ in area. The growth rate splits into an edge-entry probability and an edge-occupation ratio that stay constant for the large majority of rules, together with an edge that scales close to $A^{\frac{1}{2}}$; rules that violate the cube root are those whose edge occupation drifts (correlation of α with γ_κ 0.78 over blob rules); this is bookkeeping through Equation 2, not a cause. The result is empirical and stops short of a proof, and the quantitative values of the constants are explained only for c (by a rule-letter model) and not for κ or β. We did measure the factors for the rotor with arrows initialised with a northward bias (`notes/frontier-2026-10-13-results.md`, C2b): κ rose from 2.01 to 2.20 at bias 0.15 while the width exponent stayed at 0.32–0.33, until the cube root broke between bias 0.25 and 0.30; randomising the arrow with probability ε raised κ up to about 3-fold and destroyed the cube root (Arm B). Neither intervention explains the ants' rule-specific κ. A natural next step is to extend the census beyond $n = 12$ with a stated-in-advance prediction, and to look for a rule-level quantity that, unlike the letters, the necklace and the edge-layer shape (all tested and failed), orders κ.

The future-window forecast adds evidence beyond the same-window identities: early transport budgets predict later increments substantially better than an other-rule baseline in the typical replicated run. Rare near-stalls defeat the forecast, and the physical origin of persistent rule-level transport remains open.

11 Availability and statements

Data and code availability. [PLACEHOLDER: repository URL and archived DOI (e.g. Zenodo) to be added once the working tree is committed.] The Rust simulator, the Python analysis scripts, the generated data files, `research-outputs/cube-root-growth-paper/compute_numbers.py` (every number in this text) and `research-outputs/cube-root-growth-paper/figures.py` are in the repository; the exact commands are in `research-outputs/cube-root-growth-paper/README.md` and `notes/`. At the time of writing the tree is uncommitted, so no commit hash is quoted.

Use of AI tools. The code, analyses, and draft of this paper were produced with substantial assistance from Claude (Anthropic) via Claude Code and Google Gemini via Antigravity, under the author's direction. Codex (OpenAI) also assisted with research planning, hypothesis development, and critical

review of proposed experimental designs. The author is responsible for the content and has reviewed the results. [Check wording against the target venue's policy.]

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